

Data Flow Diagram

Date	15 March 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval/Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

Proposed DFD Components for Your Project

Since the DFD requires a visual diagram, here are the defined components and flows for your **Credit Card Approval Prediction** system so you can easily draw it or build it in a diagram tool:

1. External Entities (Ovals):

- **Applicant (User):** Submits their application details.
- **Bank Agent (Admin):** Reviews predictions and dashboard.

2. Processes (Numbered Rectangles):

- **1. Application Handling:** Receives and validates raw user inputs.
- **2. Data Preprocessing:** Cleans missing values, encodes categories, and scales data.
- **3. ML Prediction:** Runs the classification model (e.g., XGBoost) to predict approval probability.
- **4. XAI Explanation:** Uses SHAP values to generate the reason for the decision.

3. Data Stores (Solid Rectangles):

- **Applicant Database:** Stores incoming raw applications.
- **Prediction Records:** Stores the output predictions and XAI explanations for future reference.

4. Data Flows (Arrows):

- **Applicant → (1) Application Handling:** "Application Form Data"
- **(1) Application Handling → Applicant Database:** "Validated Raw Data"
- **Applicant Database → (2) Data Preprocessing:** "Stored Application Data"
- **(2) Data Preprocessing → (3) ML Prediction:** "Cleaned & Scaled Features"
- **(3) ML Prediction → (4) XAI Explanation:** "Approval Status & Prediction Probability"
- **(3) ML Prediction → Prediction Records:** "Prediction Result"
- **(4) XAI Explanation → Applicant / Bank Agent:** "Approval Status + Detailed Explanation"